

Normalization in Deep Learning

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Why Normalization?

- Stabilizes training of deep neural networks
- Reduces internal covariate shift
- Improves convergence speed
- Enables higher learning rates

Local Response Normalization (LRN)

Used in: AlexNet (2012)

- Normalizes activations across neighboring channels
- Inspired by biological lateral inhibition
- Applied after ReLU

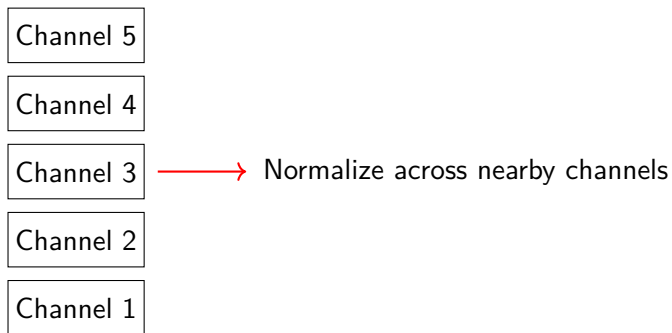
$$b_{x,y}^i = \frac{a_{x,y}^i}{\left(k + \alpha \sum_{j=i-n/2}^{i+n/2} (a_{x,y}^j)^2\right)^\beta}$$

where $a_{x,y}^i$ and $b_{x,y}^i$ denote the input and normalized activations at spatial location (x, y) in channel i , n is the number of neighboring channels, and k , α , and β are hyperparameters.

Limitations:

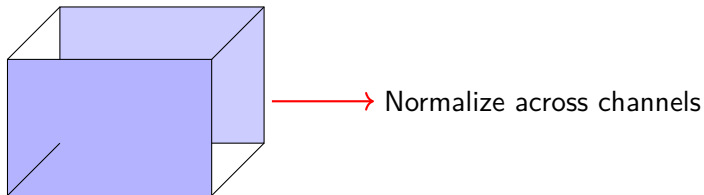
- Computationally expensive
- Limited benefit in deep networks
- Rarely used today

LRN: Channel-wise Normalization



Key idea: Competition among neighboring feature maps at the same spatial location.

LRN: Channel-wise Normalization



LRN: Competition among nearby channels at the same spatial location. **Same pixel (x,y), multiple channels → compete.**

Batch Normalization (BN)

Introduced: 2015

- Normalizes activations across the mini-batch
- Reduces internal covariate shift
- Enables faster and more stable training

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

- x : input activation, $\mu_B = \frac{1}{m} \sum_{i=1}^m x_i$: batch mean
- $\sigma_B^2 = \frac{1}{m} \sum_{i=1}^m (x_i - \mu_B)^2$: batch variance, m : batch size
- ϵ : small constant for numerical stability, γ, β : learnable scale and shift parameters

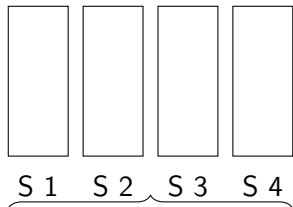
Pros:

- Faster convergence, Prevents exploding/vanishing gradients
- Regularization effect, Allows higher learning rates

Cons:

- Depends on batch size

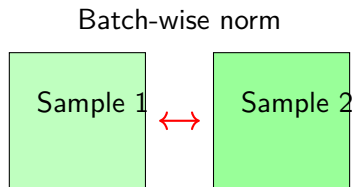
Batch Normalization



Normalize across batch

Key idea: Mean and variance computed over mini-batch.

Batch Normalization



BatchNorm: Same channel normalized across batch samples. **Different images** → **same channel** → **shared statistics**.

Layer Normalization (LN)

- Normalizes across features within a layer and Independent of batch size
- Same normalization during training and inference

$$\mu_L = \frac{1}{H} \sum_{j=1}^H x_j \quad (1)$$

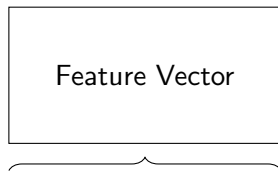
$$\sigma_L^2 = \frac{1}{H} \sum_{j=1}^H (x_j - \mu_L)^2 \quad (2)$$

$$\hat{x}_j = \frac{x_j - \mu_L}{\sqrt{\sigma_L^2 + \epsilon}} \quad (3)$$

$$y_j = \gamma \hat{x}_j + \beta \quad (4)$$

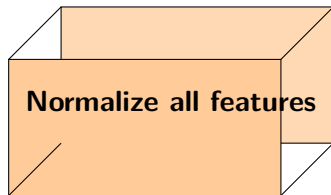
H is the number of features, ϵ is a small constant for numerical stability, and γ, β are learnable scale and shift parameters. **Advantages:**

- Stable for small batches and Ideal for sequence models, **Common in: Transformers, RNNs**



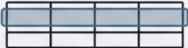
Normalize across features

Key idea: Independent of batch size; ideal for Transformers.



LayerNorm: Normalizes across all channels and spatial features. **One sample → normalize everything inside it.**

Batch Norm vs. Layer Norm

Batch Norm	Layer Norm
 Vertical (Across Batch)	 Horizontal (Across Features)
• Axis: Vertical (Across Batch) • Dependency: Dependent on Batch Size • Failure Mode: Breaks with variable sequence lengths (Padding pollution) • Use Case: Computer Vision (CNNs)	• Axis: Horizontal (Across Features) • Dependency: Independent (Per Sample) • Strength: Robust to Padding and variable lengths • Use Case: NLP (Transformers / RNNs)

Group Normalization (GN)

Introduced: 2018

- Channels divided into groups, Normalization applied within each group and Does not rely on batch statistics

Benefits:

- Performs well with small batch sizes and Effective for detection and segmentation tasks

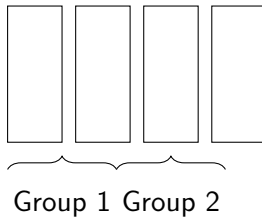
$$\mu_g = \frac{1}{m} \sum_{i=1}^m x_i \quad (5)$$

$$\sigma_g^2 = \frac{1}{m} \sum_{i=1}^m (x_i - \mu_g)^2 \quad (6)$$

$$\hat{x}_i = \frac{x_i - \mu_g}{\sqrt{\sigma_g^2 + \epsilon}} \quad (7)$$

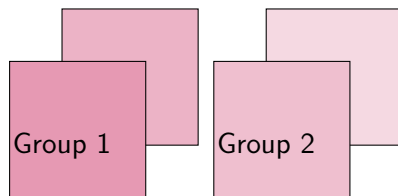
$$y_i = \gamma \hat{x}_i + \beta \quad (8)$$

Group Normalization



Key idea: Channels divided into groups; batch-size independent.

Group Normalization



GroupNorm: Normalize within channel groups (batch-size independent). **Split channels** → **normalize per group.**

Instance Normalization (IN)

Instance Normalization normalizes activations per instance and per channel, computing statistics independently for each sample. It is especially effective in image generation and style transfer, where instance-specific contrast matters more than batch statistics.

$$\mu_{n,c} = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W x_{n,c,h,w} \quad (9)$$

$$\sigma_{n,c}^2 = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W (x_{n,c,h,w} - \mu_{n,c})^2 \quad (10)$$

$$\hat{x}_{n,c,h,w} = \frac{x_{n,c,h,w} - \mu_{n,c}}{\sqrt{\sigma_{n,c}^2 + \epsilon}} \quad (11)$$

$$y_{n,c,h,w} = \gamma_c \hat{x}_{n,c,h,w} + \beta_c \quad (12)$$

Instance Normalization (IN)

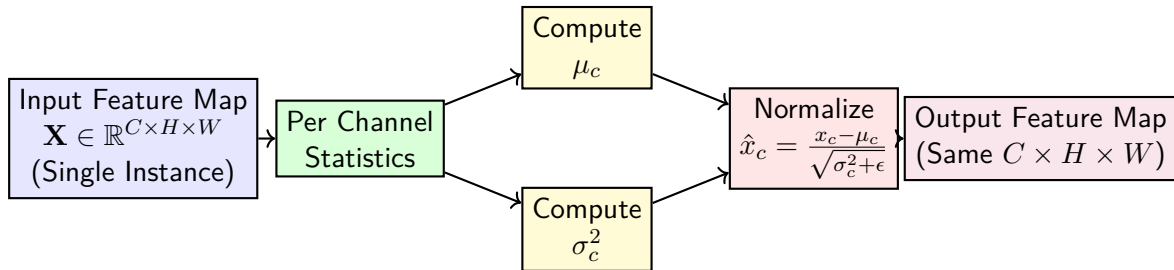


Figure: Instance Normalization computes mean and variance independently for each channel of each individual instance, preserving instance-specific contrast information.

Normalization: Visual Intuition

- **LRN:** Compete with neighboring channels or Depth-wise competition
- **BatchNorm:** Normalize across batch samples
- **LayerNorm:** Normalize across features
- **GroupNorm:** Normalize within channel groups

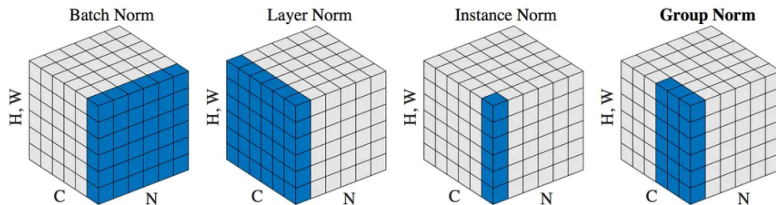


Figure: Normalization

Differences between normalization techniques

Table: Key differences between normalization techniques

Method	Normalized Over	Batch Dependent	Typical Use
Batch Normalization	Batch + Spatial	Yes	CNNs
Layer Normalization	All Features	No	Transformers
Group Normalization	Channel Groups	No	Detection
Instance Normalization	Spatial (per channel)	No	Style Transfer

Comparison Summary

Method	Norm Over	Batch Dep.	Used Today
LRN	Channels	No	Rare
BatchNorm	Batch	Yes	Yes
LayerNorm	Features	No	Yes
GroupNorm	Channel Groups	No	Yes